

Locally Thermal from Global Athermality

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Abstract

We study a symmetric bipartite quantum system AA' at inverse temperature β with identical local Hamiltonians $H_A = H_{A'} =: H$ and local Gibbs state γ . A bipartite state $\rho_{AA'}$ is *locally thermal* (LT) when $\text{Tr}_{A'} \rho_{AA'} = \gamma$ and $\text{Tr}_A \rho_{AA'} = \gamma$. This removes all local athermality, so any deviation from the global equilibrium state $\Gamma := \gamma \otimes \gamma$ is stored entirely in correlations. The project asks three linked questions: what is the structure of the LT set, how do correlations evolve under global Gibbs-preserving operations, and how does the convertibility hierarchy change when one restricts to local Gibbs-preserving maps?

Analytically, the main target is the full two-qubit LT problem. We show that LT is an affine slice of the positive semidefinite cone, derive its dimension, obtain the full 9-parameter two-qubit LT normal form, and reduce generic two-qubit positivity to an explicit Schur-complement criterion. We also analyse the diagonal- T slice, exact LT-ray feasibility bounds, and the commuting qutrit subclass, which forms a 4-dimensional transportation polytope inside the full 64-dimensional qutrit LT body.

Operationally, the identity $D(\rho||\Gamma) = I(A : B)_\rho$ on LT is used as our core reference. Its consequence is that Gibbs-preserving monotonicity of relative entropy becomes mutual information monotonicity on LT. We also give an explicit local Gibbs Preserving operation on our thermo field double (TFD) to dephased chain, formulate the relevant semidefinite programs, and interpret numerical results on Positive Partial Transpose (PPT) structure, global versus-local convertibility, and higher dimensional commuting LT samples.

1 Introduction

1.1 Problem setting and motivation

Modern thermodynamics treats information as physical: correlations, uncertainty, and coherence affect energetic value, work extraction, and state convertibility [1, 2, 3, 4]. The present project isolates a regime where all *local* nonequilibrium structure has already been removed. We consider a symmetric bipartite system AA' with identical local Hamiltonians $H_A = H_{A'} =: H$, local Gibbs state $\gamma = e^{-\beta H}/Z$, and global equilibrium state $\Gamma := \gamma \otimes \gamma$. A bipartite state $\rho_{AA'}$ is *locally thermal* (LT) when both reduced states are exactly thermal, $\text{Tr}_{A'} \rho_{AA'} = \gamma$ and $\text{Tr}_A \rho_{AA'} = \gamma$.

This constraint removes all local athermality by construction. If such a state still differs from Γ , that deviation can no longer be stored in local populations or local coherences; it must be encoded globally, through correlations. The project therefore turns a thermodynamic question into a geometric and operational one: what is the structure of states with fixed thermal marginals, and what resource remains once local equilibrium is enforced exactly?

Why this is important. The LT constraint simplifies the thermodynamic properties, but it does not make the state space easy. Fixing both marginals removes local freedom while leaving a high dimensional positivity problem for the remaining correlation sector. Even in the first genuinely nontrivial symmetric case, $d = 2$, the LT set is already 9 dimensional, and positivity couples the correlation parameters in a nonlinear way. This is why the project is not merely about exhibiting special examples: the main goal is to understand the full constrained geometry and then ask how that geometry interacts with Gibbs preserving dynamics.

In particular, we ask the following questions:

1. What is the structure of a bipartite quantum state ρ_{AB} if its reduced states ρ_A and ρ_B are thermal?
2. How do correlations in ρ_{AB} evolve under global thermalisation processes, such as Gibbs preserving operations that maintain the global thermal state?
3. How do correlations in ρ_{AB} change when only local Gibbs-preserving operations (those that preserve the local thermal states) are applied?

1.2 Project scope and internal logic

The report is organised around a deliberate separation of scales. For symmetric qubits, $\dim_{\mathbb{R}} \text{LT} = (2^2 - 1)^2 = 9$, which is the first nontrivial case where a near complete analytic description of the full LT body is still realistic. The main target of the project is not merely to analyse a few slices, but to describe the full two-qubit LT geometry as explicitly as possible.

Around that main target sit two supporting layers. First, there are controlled subfamilies such as the diagonal T slice and one parameter LT rays. These are analytically tractable, they admit exact positivity and separability criteria, and they provide stable numerical probes for convertibility. Second, there is the higher dimensional qutrit extension. For $d = 3$, the full symmetric LT set

already has real dimension $(3^2 - 1)^2 = 64$, so a qubit style complete characterization is unrealistic. The correct qutrit contribution is therefore not “solve the full qutrit problem” but rather identify analytically meaningful subclasses. The commuting qutrit LT family is the natural case because it reduces exactly to a transportation polytope.

This gives the project the following structure:

LT geometry $\rightarrow$ two-qubit characterization $\rightarrow$ slices/rays & convertibility $\rightarrow$ qutrit subclass.

1.3 Main contributions

The main analytic contributions are the following: LT is identified as an affine slice of the positive semidefinite cone and hence a spectrahedron [5, 6]. Its real dimension is shown to be $(d_A^2 - 1)(d_B^2 - 1)$. In the symmetric two-qubit case every LT state admits an explicit 9 parameter matrix normal form, and generic positivity reduces to an explicit Schur complement criterion. The diagonal T slice yields exact positivity and separability inequalities, while LT ray feasibility can be characterized exactly by whitening with respect to Γ .

The main thermodynamic observation is that on the LT manifold the free energy monotone $D(\rho||\Gamma)$ coincides exactly with the mutual information $I(A:A')_\rho$ [7, 8]. Since this follows immediately once the marginals are fixed, its main use here is as a tool: GP monotonicity of relative entropy becomes mutual information monotonicity on LT. A short calculation also shows that LT is closed under local GP maps, and the thermofield-double (TFD) to dephased family gives an explicit ordered chain under local GP. In dimension $d = 3$, the commuting LT subclass is shown to be a 4-dimensional transportation polytope with exact mutual information range $0 \leq I \leq H(g)$ [9]. Here I denotes mutual information, $g = (g_1, g_2, g_3)$ is the Gibbs probability vector of $\gamma = \text{diag}(g_1, g_2, g_3)$, $H(g) := -\sum_i g_i \log g_i$ is its Shannon entropy, and for a joint population matrix $P = (p_{ij})$ we write $H(P) := -\sum_{i,j} p_{ij} \log p_{ij}$.

The numerical part of the report is designed to reflect those analytic structures rather than replace them. The figures illustrate the LT identity $D = I$ where useful, classify Positive Partial Transpose (PPT) versus non PPT LT samples, compare global, local, and PPT relaxed convertibility graphs, study the detailed Pauli XX ray, and sample the commuting qutrit face. Throughout, the emphasis is on interpreting these outputs physically: when a plot shows inward global convertibility, sign-flip symmetries, or a low correlation cluster of PPT states, the point is not just the picture itself but what feature of the LT geometry or GP constraint explains it.

Transportation polytopes. A transportation polytope is the convex set of nonnegative matrices with prescribed row and column sums. In the present commuting LT setting, the matrix $P = (p_{ij})$ of joint energy populations satisfies

$$\sum_j p_{ij} = g_i, \quad \sum_i p_{ij} = g_j, \quad p_{ij} \geq 0,$$

so the commuting subclass is exactly the transportation polytope $T(g, g)$ [9]. Because the margins are fixed and the entries are nonnegative, this set is bounded; hence it is a polytope rather than an unbounded polyhedron.

1.4 Context in the existing literature

The broader resource theoretic approach to quantum thermodynamics is now well developed: free energy monotones, Gibbs preserving maps, thermal operations, and correlation based resource questions have been studied extensively in finite dimensions [1, 2, 3, 4]. At the same time, the general geometry of bipartite states with fixed thermal marginals is much less explicit. Even for two qubits, fixing both marginals leaves a genuinely nonlinear positivity problem in the correlation sector.

This project sits exactly at that gap. Rather than starting from asymptotic thermodynamic laws or from highly specialised examples, it treats locally thermal states as a finite dimensional constrained state space in their own right and asks for an explicit structural description. The point is therefore not just to produce examples of correlated locally thermal states, but to develop a framework in which geometry, thermodynamic monotones, and convertibility can be analysed together. In qubits this means an explicit characterisation of the full LT body; in qutrits it means identifying analytically tractable subclasses rather than claiming an unrealistic full classification. Operationally, this also separates two levels of dynamical control. Under global Gibbs preserving maps one asks what transformations are compatible with preservation of the bipartite Gibbs state. Under local Gibbs preserving maps one asks what survives once the dynamics is forced to factorise. That contrast is important both physically and computationally, and it is one of the main reasons the project naturally combines analytic work with semidefinite programming methods.

2 Background

2.1 Quantum states, channels, and correlations

A finite dimensional quantum state is a density operator $\rho \in \mathcal{D}(\mathcal{H})$ with $\rho = \rho^\dagger$, $\rho \succeq 0$, and $\text{Tr } \rho = 1$. For a bipartite state ρ_{AB} the marginals are $\rho_A = \text{Tr}_B \rho_{AB}$ and $\rho_B = \text{Tr}_A \rho_{AB}$. Physical transformations are completely positive trace preserving (CPTP) maps. We use the Choi representation when convenient, because Gibbs preserving (GP) convertibility becomes a semidefinite feasibility problem in the Choi matrix [10, 11, 6].

Three correlation classes are relevant throughout. Product states have the form $\rho_{AB} = \rho_A \otimes \rho_B$. Separable states are convex mixtures $\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}$. Entangled states are those that are not separable. In the two qubit numerical sections this distinction becomes operational because positivity under partial transpose (PPT) is equivalent to separability in 2×2 dimensions [12, 13].

2.2 Entropy, relative entropy, and Gibbs preserving operations

The von Neumann entropy is $S(\rho) := -\text{Tr}(\rho \log \rho)$. The quantum relative entropy is $D(\rho \parallel \sigma) := \text{Tr}[\rho(\log \rho - \log \sigma)]$ when $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$. For a bipartite state ρ_{AB} the mutual information is

$$I(A:B)_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) = D(\rho_{AB} \parallel \rho_A \otimes \rho_B),$$

which makes it the natural total correlation measure [7]. Relative entropy is contractive under CPTP maps, so $D(\rho \parallel \sigma) \geq D(\Phi(\rho) \parallel \Phi(\sigma))$ for every channel Φ .

Given a Hamiltonian H and inverse temperature β which we set to 1, the Gibbs state is $\gamma = e^{-\beta H}/Z$. Relative entropy to γ has the free energy form $D(\rho \parallel \gamma) = \beta(F(\rho) - F(\gamma))$, so it measures nonequilibrium free energy content relative to thermal equilibrium [8, 2]. A channel Φ is *Gibbs-preserving* when $\Phi(\gamma) = \gamma$. By data processing, this implies monotonicity of $D(\cdot \parallel \gamma)$ for all inputs. Conversely, if $D(\Phi(\rho) \parallel \gamma) \leq D(\rho \parallel \gamma)$ for all ρ , then setting $\rho = \gamma$ gives $D(\Phi(\gamma) \parallel \gamma) \leq 0$, hence $\Phi(\gamma) = \gamma$. We use this GP framework because it gives a broad thermodynamic class while remaining directly compatible with the SDP methods used later.

Three structural facts drive the rest of the report. First, the thermofield double vector is pure and locally thermal. The vector

$$|\tilde{\gamma}\rangle_{AA'} = \sum_i \sqrt{g_i} |i\rangle_A \otimes |i\rangle_{A'}$$

is the canonical purification of the Gibbs state γ , often called the thermofield double (TFD) state [7]. In the nondegenerate case it is the unique pure LT state up to local phases: if $|\psi\rangle$ is pure and $\text{Tr}_{A'} |\psi\rangle\langle\psi| = \gamma$, then its Schmidt coefficients are $\{\sqrt{g_i}\}$. Nondegeneracy fixes the Schmidt basis to the energy basis up to phases, and the equal marginal constraint forces the right Schmidt basis to coincide with the left one up to the same phase freedom. Degeneracies enlarge this freedom to local unitaries acting inside degenerate eigenspaces. This gives a distinguished extremal LT state and a natural reference family for convertibility.

Second, on the LT manifold one has the exact identity

$$D(\rho \| \gamma_A \otimes \gamma_B) = I(A:B)_\rho.$$

This is immediate from the general formula $I(A:B)_\rho = D(\rho_{AB} \| \rho_A \otimes \rho_B)$ after substituting the LT marginal constraints [7]. We use this mainly as a corollary: once both marginals are fixed, any GP monotonicity statement for $D(\cdot \| \gamma_A \otimes \gamma_B)$ becomes a monotonicity statement for mutual information on LT.

Third, LT is closed under local GP operations. If $\Phi_A(\gamma_A) = \gamma_A$ and $\Phi_B(\gamma_B) = \gamma_B$, then for every $\rho \in \text{LT}(\gamma_A, \gamma_B)$ one has $(\Phi_A \otimes \Phi_B)(\rho) \in \text{LT}(\gamma_A, \gamma_B)$ because the marginal constraints are preserved under partial trace. Consequently, mutual information cannot increase under local GP maps when both input and output remain in LT.

2.3 Why relative entropy equals free energy excess

A basic identity used throughout the project is

$$D(\rho \| \gamma) = \beta(F(\rho) - F(\gamma)),$$

where

$$F(\rho) := \text{Tr}(\rho H) - \beta^{-1} S(\rho), \quad S(\rho) := -\text{Tr}(\rho \log \rho),$$

and $\gamma = e^{-\beta H}/Z$ is the Gibbs state [8, 2]. Since $\log \gamma = -\beta H - \log Z$,

$$D(\rho \| \gamma) = \text{Tr}[\rho(\log \rho - \log \gamma)] = -S(\rho) + \beta \text{Tr}(\rho H) + \log Z = \beta F(\rho) + \log Z.$$

Using $F(\gamma) = -\beta^{-1} \log Z$, equivalently $\log Z = -\beta F(\gamma)$, gives

$$D(\rho \| \gamma) = \beta(F(\rho) - F(\gamma)).$$

Thus relative entropy to γ is exactly the nonequilibrium free energy excess.

2.4 Locally thermal states as an affine positive body

States with thermal marginals are physically meaningful because they isolate correlation as the only remaining nonequilibrium resource once each subsystem is already in local equilibrium. In that sense they give a finite-dimensional setting in which the thermodynamic role of correlations can be studied without contamination from local athermality [3, 4, 1]. For fixed local Gibbs states γ_A and γ_B , define

$$\text{LT}(\gamma_A, \gamma_B) := \{\rho_{AB} : \text{Tr}_B \rho_{AB} = \gamma_A, \text{Tr}_A \rho_{AB} = \gamma_B\}.$$

In the symmetric case $\gamma_A = \gamma_B = \gamma$, we write simply $\text{LT}(\gamma)$. If $\Gamma := \gamma_A \otimes \gamma_B$, then every LT state can be written as $\rho = \Gamma + C$, where $C = C^\dagger$, $\text{Tr}_A C = \text{Tr}_B C = 0$, and $\Gamma + C \succeq 0$. Hence

$$\text{LT}(\gamma_A, \gamma_B) = (\Gamma + \mathcal{Z}) \cap \text{PSD}, \quad \mathcal{Z} := \{C = C^\dagger : \text{Tr}_A C = \text{Tr}_B C = 0\}.$$

This immediately shows that LT is a spectrahedron, i.e. an affine slice of the positive semidefinite cone [5, 6]. The affine constraints are linear; positivity is the only nonlinear ingredient.

Dimension counting is therefore clean. A bipartite density operator on $\mathcal{H}_A \otimes \mathcal{H}_B$ has $(d_A d_B)^2 - 1$ real degrees of freedom. Fixing the A marginal imposes $d_A^2 - 1$ independent real constraints, and fixing the B marginal imposes $d_B^2 - 1$ more. Thus

$$\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1).$$

For symmetric qubits this gives $\dim_{\mathbb{R}} \text{LT} = 9$; for symmetric qutrits it gives $\dim_{\mathbb{R}} \text{LT} = 64$. This scale separation is one of the main reasons the report is organized around a full qubit treatment and a structured qutrit subclass rather than an attempted full qutrit classification.

In an orthonormal traceless operator basis, every $C \in \mathcal{Z}$ expands purely through correlation terms. Thus LT can be viewed as a real correlation-tensor space cut by positivity. This picture is useful both analytically and numerically: analytically because it makes the dimension count transparent, and numerically because it suggests natural parameterizations of special families.

3 Analytic results

3.1 The Minimum Model - Two Qubit Case

For symmetric qubits the LT set has real dimension 9. This is the first case where one can still hope to describe the full LT body rather than only selected low-dimensional sections. It is therefore the first genuinely nontrivial setting in which the project can move from examples to a full structural result. Throughout this section we write

$$\gamma = \begin{pmatrix} g & 0 \\ 0 & 1 - g \end{pmatrix}, \quad \frac{1}{2} \leq g < 1,$$

with $g \neq \frac{1}{2}$ the generic finite-temperature regime and $g = \frac{1}{2}$ the infinite-temperature limit.

3.1.1 Exact canonical matrix form

Proposition 3.1 (Canonical 9-parameter two-qubit LT form). *A two-qubit state ρ is locally thermal with both marginals equal to γ if and only if, in the product energy basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$,*

$$\rho = \begin{pmatrix} g - a & x & y & z \\ x^* & a & w & -y \\ y^* & w^* & a & -x \\ z^* & -y^* & -x^* & 1 - g - a \end{pmatrix}, \quad a \in \mathbb{R}, \ x, y, z, w \in \mathbb{C}, \quad (1)$$

with the sole remaining condition $\rho \succeq 0$.

Proof sketch. Start from a general Hermitian 4×4 matrix in the product energy basis. Hermiticity gives four real diagonal entries and six complex off-diagonal entries, hence 16 real parameters, or 15 after the trace-one condition. Imposing both marginal constraints gives six independent real linear equations: three from each marginal, with the trace overlap counted only once. Solving them row by row yields the sign pattern in Eq. (1), with diagonal entries $(g - a, a, a, 1 - g - a)$ and off-diagonal relations $(\rho_{24}, \rho_{34}) = (-\rho_{13}, -\rho_{12})$. The remaining free data are one real parameter a and four complex parameters x, y, z, w , i.e. 9 real degrees of freedom.

The marginal constraints remove six independent real degrees of freedom from the original 15, leaving the expected 9. Positivity immediately gives $0 \leq a \leq \min(g, 1 - g)$ together with

$$|x|^2, |y|^2 \leq a(g - a), \quad |w|^2 \leq a(1 - g - a), \quad |z|^2 \leq (g - a)(1 - g - a),$$

but these bounds are not sufficient. The real difficulty is the coupled positivity geometry of the full off-diagonal sector.

For nondegenerate γ , the local unitaries that preserve the Gibbs state are diagonal phase rotations. They preserve LT and allow a gauge in which $x, y \in \mathbb{R}_{\geq 0}$. Writing $\alpha := g - a$ and $\delta := 1 - g - a$,

one obtains

$$\rho = \begin{pmatrix} \alpha & x & y & z \\ x & a & w & -y \\ y & w^* & a & -x \\ z^* & -y & -x & \delta \end{pmatrix}, \quad x, y \in \mathbb{R}_{\geq 0}, \quad z, w \in \mathbb{C}. \quad (2)$$

3.1.2 Schur-complement reduction of the full positivity problem

Partition (2) into 2×2 blocks,

$$\rho = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A = \begin{pmatrix} \alpha & x \\ x & a \end{pmatrix}, \quad B = \begin{pmatrix} y & z \\ w & -y \end{pmatrix}, \quad D = \begin{pmatrix} a & -x \\ -x & \delta \end{pmatrix},$$

and let $\Delta := \det A = a\alpha - x^2$.

Theorem 3.2 (Generic positivity criterion for two-qubit LT states). *Let ρ be of the form (2), and assume $\Delta > 0$. Then $\rho \succeq 0$ if and only if $0 \leq a \leq 1 - g$ and the Schur complement*

$$N := D - B^\dagger A^{-1} B = \begin{pmatrix} N_{11} & N_{12} \\ N_{12}^* & N_{22} \end{pmatrix}$$

satisfies $N_{11} \geq 0$, $N_{22} \geq 0$, and $N_{11}N_{22} \geq |N_{12}|^2$, where

$$\begin{aligned} N_{11} &= a - \frac{ay^2 + \alpha|w|^2 - 2xy \Re(w)}{\Delta}, \\ N_{22} &= \delta - \frac{\alpha y^2 + a|z|^2 + 2xy \Re(z)}{\Delta}, \\ N_{12} &= \frac{-x\Delta - ayz + \alpha y w^* + x z w^* - xy^2}{\Delta}. \end{aligned} \quad (3)$$

Proof sketch. For $\Delta > 0$, the block A is positive and invertible. By the Schur-complement criterion, $\rho \succeq 0$ is equivalent to $A \succeq 0$ and $D - B^\dagger A^{-1} B \succeq 0$. Using

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} a & -x \\ -x & \alpha \end{pmatrix}$$

and substituting the explicit blocks gives the displayed formulas for N_{11} , N_{22} , and N_{12} . Since the Schur complement is Hermitian 2×2 , its positivity is equivalent to nonnegative diagonal entries together with $\det N \geq 0$, i.e. $N_{11}N_{22} \geq |N_{12}|^2$. The singular boundary $\Delta = 0$ requires the pseudoinverse form of the criterion and is treated separately in the appendix.

This theorem reduces the generic interior of the full 9-dimensional LT body to explicit inequalities and makes the qubit LT set a concrete semi-algebraic object rather than an abstract affine slice.

3.2 Structured subfamilies

3.2.1 Classical diagonal qubit LT line

Restricting to commuting two-qubit LT states gives

$$\rho_{\text{cl}} = \text{diag}(q, g - q, g - q, 1 - 2g + q), \quad \max(0, 2g - 1) \leq q \leq g.$$

Equivalently,

$$P = \begin{pmatrix} q & g - q \\ g - q & 1 - 2g + q \end{pmatrix}$$

is exactly the 2×2 transportation polytope with margins $(g, 1 - g)$. This family is the simplest LT baseline: it already contains nontrivial classical correlation, but no coherence.

3.2.2 Pauli tensor parameterization and the diagonal- T slice

Every two-qubit LT state can be written as

$$\rho = \frac{1}{4} \left(\mathbb{I} \otimes \mathbb{I} + z_\gamma (\sigma_z \otimes \mathbb{I} + \mathbb{I} \otimes \sigma_z) + \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j \right), \quad (4)$$

where $z_\gamma := 2g - 1$ and $T \in \mathbb{R}^{3 \times 3}$. Equation (4) is the LT-specialised form of the standard Bloch/Fano decomposition of a two-qubit state [7]. In full generality one writes

$$\rho = \frac{1}{4} \left(I \otimes I + \sum_{i=x,y,z} r_i \sigma_i \otimes I + \sum_{i=x,y,z} s_i I \otimes \sigma_i + \sum_{i,j=x,y,z} T_{ij} \sigma_i \otimes \sigma_j \right).$$

In the energy basis, the LT marginal constraint

$$\rho_A = \rho_B = \gamma = \frac{1}{2} (I + z_\gamma \sigma_z), \quad z_\gamma = 2g - 1,$$

forces

$$r = s = (0, 0, z_\gamma).$$

That is why only the local σ_z -term survives in Eq. (4): local x - or y -components would generate local coherences and hence violate local thermality.

Restricting to $T = \text{diag}(t_x, t_y, t_z)$ gives a three-parameter X -state family [14, 15] with exact positivity conditions

$$\begin{aligned} (4g^2 + t_z)(4(1 - g)^2 + t_z) &\geq (t_x - t_y)^2, \\ t_z + |t_x + t_y| &\leq 4g(1 - g). \end{aligned} \quad (5)$$

At $g = \frac{1}{2}$ these reduce to the Bell-diagonal tetrahedron $1 \pm t_x \pm t_y \pm t_z \geq 0$.

Remark 3.3. The inequalities in Eq. (5) are the diagonal- T specialisation of the general positivity problem. Equivalently, once $T = \text{diag}(t_x, t_y, t_z)$, the LT state becomes an X -state in the product energy basis, and positivity reduces to positivity of the two relevant 2×2 principal blocks [14, 15].

3.2.3 The PPT and X-state structure

For a bipartite state ρ_{AB} , the partial transpose with respect to B is obtained by transposing only the B -indices in a fixed product basis. A state is Positive Partial Transpose *PPT* when $\rho^{T_B} \succeq 0$. By the Peres–Horodecki criterion, PPT is equivalent to separability in 2×2 and 2×3 dimensions [12, 13]. In the present two-qubit LT setting this makes PPT an exact separability test and a convenient SDP-compatible constraint.

When the Pauli correlation tensor is restricted to diagonal form, the LT state becomes an *X*-state in the product energy basis, with support only on diagonal and anti-diagonal entries [14, 15]. This diagonal- T slice is the simplest coherent deformation of the classical commuting LT line and will serve as the main analytically tractable subfamily discussed below.

3.2.4 Exact separability criterion on the diagonal- T slice

In 2×2 dimensions, PPT is equivalent to separability [12, 13]. Partial transpose swaps the two off-diagonal *X*-state couplings, so the known *X*-state PPT inequalities [14, 15] become

$$\begin{aligned} (4g^2 + t_z)(4(1 - g)^2 + t_z) &\geq (t_x + t_y)^2, \\ t_z + |t_x - t_y| &\leq 4g(1 - g). \end{aligned} \tag{6}$$

Hence a diagonal- T LT state is separable if and only if it satisfies both (5) and (6). This yields the exact hierarchy

$$\text{classical LT} \subset \text{separable LT} \subset \text{LT}.$$

3.2.5 Exact positivity bounds on LT rays

Recall that every LT state can be written where $C = C^\dagger$ is traceless and has vanishing marginals,

$$\rho = \Gamma + C, \quad \Gamma := \gamma_A \otimes \gamma_B, \quad \text{Tr}_A C = 0, \quad \text{Tr}_B C = 0.$$

The affine space of such centred correlation operators is

$$\mathcal{Z} := \{ C = C^\dagger : \text{Tr}_A C = 0, \text{Tr}_B C = 0 \}.$$

The point of this decomposition is conceptual as well as algebraic: C contains exactly the departure from the product Gibbs state once all local thermal content has been removed.

Given any Hermitian direction $C_0 \in \mathcal{Z}$, we consider the one-parameter family

$$\rho(p) = \Gamma + pC_0.$$

Calling C_0 a *zero-marginal direction* simply means that moving along it does not change either reduced state to first order or at finite p ; the local marginals remain fixed because

$$\text{Tr}_A C_0 = \text{Tr}_B C_0 = 0.$$

So the ray stays inside the affine LT space and only positivity decides which values of p are actually physical.

Whitening by the strictly positive background state Γ gives

$$\tilde{C}_0 := \Gamma^{-1/2} C_0 \Gamma^{-1/2}, \quad \rho(p) \succeq 0 \iff I + p\tilde{C}_0 \succeq 0.$$

If $\{\lambda_k\}$ are the eigenvalues of $\tilde{C}_0$, the exact feasible interval is

$$p \in \left[\max_{\lambda_k > 0} \left(-\frac{1}{\lambda_k} \right), \min_{\lambda_k < 0} \left(-\frac{1}{\lambda_k} \right) \right].$$

This is why LT rays are analytically clean: the full positivity problem collapses to a spectral bound for the whitened centred-correlation direction.

3.3 Qutrit LT: full body versus tractable subclass

For $d = 3$, the full symmetric LT set has real dimension $(3^2 - 1)^2 = 64$. In an orthonormal traceless Hermitian basis $\{B_a\}_{a=1}^8$, any qutrit LT state has the form

$$\rho = \Gamma + \sum_{a,b=1}^8 c_{ab} B_a \otimes B_b,$$

with positivity as the only nonlinear constraint. This is exact, but not yet a usable classification. The analytically meaningful qutrit subproject is the commuting family. Writing $\gamma = \text{diag}(g_1, g_2, g_3)$, a commuting LT state is equivalent to a nonnegative matrix $P = (p_{ij})$ with row and column sums both equal to g :

$$\sum_j p_{ij} = g_i, \quad \sum_i p_{ij} = g_j.$$

Thus commuting qutrit LT is exactly the transportation polytope $\mathcal{T}(g, g)$ [9]. A convenient parameterization is

$$P = \begin{pmatrix} a & b & g_1 - a - b \\ c & d & g_2 - c - d \\ g_1 - a - c & g_2 - b - d & g_3 - g_1 + a + b + c + d \end{pmatrix},$$

subject to entrywise nonnegativity, so the dimension is manifestly 4. This is the same transportation-polytope structure introduced earlier: the row and column sums are fixed to the Gibbs margins, and only nonnegative couplings compatible with those margins are allowed.

The vector

$$g = (g_1, g_2, g_3)$$

is the Gibbs probability vector of the local qutrit thermal state $\gamma = \text{diag}(g_1, g_2, g_3)$, and $P = (p_{ij})$ is the joint population matrix of a commuting LT state. We write

$$H(g) := -\sum_i g_i \log g_i, \quad H(P) := -\sum_{i,j} p_{ij} \log p_{ij},$$

for the Shannon entropies of the marginal distribution g and the joint distribution P , respectively.

Proposition 3.4 (Exact mutual-information range in the commuting subclass). *For commuting qutrit LT states,*

$$0 \leq I(P) \leq H(g), \quad I(P) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{g_i g_j} = 2H(g) - H(P).$$

The minimum is attained at the product coupling $P_{\text{prod}} = gg^T$, and the maximum at deterministic permutation couplings compatible with the equal-marginal constraint.

3.4 Operational results: resource meaning and convertibility

By *resource meaning* we mean the identification of a quantity that measures how far an LT state lies above the product Gibbs baseline Γ and is monotone under the free operations of interest. By *convertibility* we mean the preorder induced by those free operations: given LT states ρ and σ , does there exist an allowed map that sends ρ to σ ? In this report the two operational classes are global Gibbs-preserving maps on the bipartite system and local Gibbs-preserving maps of product form.

3.4.1 Athermality equals correlation on LT

For any LT state ρ with marginals γ_A and γ_B ,

$$D(\rho \| \gamma_A \otimes \gamma_B) = I(A:B)_\rho.$$

Thus relative entropy to the product Gibbs state is exactly the mutual information on LT. The identity is immediate once the LT constraints are imposed, and its main use here is operational: every GP monotonicity statement for $D(\cdot \| \Gamma)$ becomes a mutual-information monotonicity statement on LT.

A Hermitian operator X is *positive semidefinite* (PSD), written $X \succeq 0$, when $\langle \psi | X | \psi \rangle \geq 0$ for all vectors $|\psi\rangle$, equivalently when all eigenvalues of X are nonnegative.

We use *GP* and *GPO* interchangeably as shorthand for *Gibbs-preserving (quantum) operations*: these are CPTP maps Φ satisfying $\Phi(\gamma) = \gamma$ on a single system, or $\Phi(\Gamma) = \Gamma$ on the full bipartite system, with $\Gamma = \gamma_A \otimes \gamma_B$.

Remark: LT is closed under local Gibbs-preserving maps. If $\Phi_A(\gamma_A) = \gamma_A$ and $\Phi_B(\gamma_B) = \gamma_B$, then for every $\rho \in \text{LT}(\gamma_A, \gamma_B)$,

$$(\Phi_A \otimes \Phi_B)(\rho) \in \text{LT}(\gamma_A, \gamma_B).$$

This gives a rigorous separation between global and local Gibbs-preserving dynamics: local GP can move states around inside LT, but it cannot create local athermality. Combined with $D(\rho \| \Gamma) = I(A:B)_\rho$ on LT, it follows that mutual information cannot increase under local GP maps acting within LT.

3.4.2 Explicit ordered TFD-to-dephased chain

Let $|\tilde{\gamma}\rangle = \sum_i \sqrt{g_i} |ii\rangle$, let $\tilde{\gamma} = |\tilde{\gamma}\rangle\langle\tilde{\gamma}|$, and let $\tilde{\gamma}_{\text{diag}}$ denote its full product-energy dephasing. Define

$$\rho_\lambda = (1 - \lambda)\tilde{\gamma} + \lambda\tilde{\gamma}_{\text{diag}}, \quad 0 \leq \lambda \leq 1,$$

and the dephasing mixture map

$$G_\mu = (1 - \mu) \text{id} + \mu \Delta, \quad 0 \leq \mu \leq 1.$$

Because local dephasing is Gibbs-preserving and factorises as $\Delta_A \otimes \Delta_B$, the map G_μ is allowed under local GP. Moreover,

$$G_\mu(\rho_\lambda) = (1 - \lambda')\tilde{\gamma} + \lambda'\tilde{\gamma}_{\text{diag}}, \quad \lambda' = \lambda + \mu(1 - \lambda).$$

Hence for every $0 \leq \lambda \leq \lambda' \leq 1$ one has $\rho_\lambda \succ_{\text{LGP}} \rho_{\lambda'}$. This gives an explicit local-GP ordered chain with monotone decrease of $D(\rho_\lambda \| \Gamma) = I(\rho_\lambda)$.

3.5 Computational methods

A semidefinite program (SDP) is an optimisation problem in which one minimises a linear objective over matrix variables subject to linear constraints and positivity constraints of the form $X \succeq 0$ [10, 11, 6]. SDPs are natural here because the core objects in the project—density matrices, Choi matrices, trace-preservation constraints, marginal constraints, and Gibbs-preservation constraints—are all linear or positive-semidefinite in the relevant variables. The analytic LT structure feeds directly into convex optimisation. LT membership is positivity together with linear marginal constraints, so it is naturally a semidefinite feasibility problem. Global GP convertibility is linear in the Choi matrix of the channel, while the local problem is harder because the tensor-product constraint on the channel is nonconvex.

SDP 1: LT membership and nearest-LT projection

LT membership asks whether a candidate state already lies in the constrained LT set:

$$\tau \succeq 0, \quad \text{Tr}_A \tau = \gamma_B, \quad \text{Tr}_B \tau = \gamma_A, \quad \text{Tr} \tau = 1.$$

To project a general state ρ onto LT, we instead minimise the trace distance to an LT state τ :

$$\min_{\tau \in \text{LT}(\gamma_A, \gamma_B)} \frac{1}{2} \|\rho - \tau\|_1.$$

Thus the optimisation is not mapping one LT state to another; it maps an arbitrary input state ρ to its closest LT approximation τ in trace distance.

SDP 2: global GP convertibility

For LT states ρ and σ , global GP convertibility asks whether there exists a CPTP map Φ on the full bipartite system such that

$$\Phi(\rho) = \sigma, \quad \Phi(\Gamma) = \Gamma, \quad \Gamma := \gamma \otimes \gamma.$$

The second constraint uses the *global* Gibbs state Γ , not the local state γ , because the channel acts on the whole bipartite system. In the Choi representation this becomes a linear feasibility problem:

$$J_\Phi \succeq 0, \quad \text{Tr}_{\text{out}} J_\Phi = I_{\text{in}},$$

together with linear equations encoding $\Phi(\rho) = \sigma$ and $\Phi(\Gamma) = \Gamma$. Hence the problem is an SDP.

SDP 3: local GP convertibility and PPT outer bounds

Local convertibility is more difficult because one asks for a product channel

$$\Phi_A \otimes \Phi_B$$

with each factor Gibbs-preserving:

$$\Phi_A(\gamma) = \gamma, \quad \Phi_B(\gamma) = \gamma, \quad (\Phi_A \otimes \Phi_B)(\rho) = \sigma.$$

The obstruction is that tensor-product channel structure is nonconvex in a single Choi variable, so the exact local problem is not a single SDP. This is why the numerics use two layers: verified local searches for genuine local edges, and PPT-based outer relaxations as necessary convex tests. Throughout the report, *verified local GP* means an edge returned in the experiment setting `local_edge_mode=verified`; the experiment guide treats this as the direct mode, while `single` is heuristic only. Accordingly, a positive verified edge is a certified output of that mode, whereas failure to certify an edge should be read as inconclusive unless the PPT outer test also rules the pair out.

The benefit of the PPT relaxation is computational: PPT constraints are semidefinite and therefore tractable. A failure of the PPT outer test rules out convertibility at that relaxation level, while a pass does *not* prove that a true local GP map exists. In other words, PPT is useful here because it gives a cheap convex outer bound on a hard nonconvex locality problem, not because it solves the local problem exactly.

3.6 Numerical results

How to read the numerical section. The analytic core of this project is the full two-qubit LT characterization, which is a 9-dimensional positivity problem rather than a collection of low-dimensional slices. The numerical figures are therefore not intended to replace the full theory. Instead, they probe representative directions and subfamilies of the LT body that isolate distinct phenomena: geometry of the feasible region, the LT relation between thermodynamic

and information-theoretic monotones, onset of entanglement, and the contrast between global and local Gibbs-preserving convertibility. In this sense, the plotted rays, slices, and benchmark families should be read as structured diagnostics of the full LT body rather than as independent stand-alone models.

Unless stated otherwise, the qubit numerics use energy gap $\varepsilon = 1$ and inverse temperature $\beta = 1$, so

$$\gamma = \frac{e^{-\beta H}}{Z} = \text{diag}\left(\frac{1}{1 + e^{-\beta\varepsilon}}, \frac{e^{-\beta\varepsilon}}{1 + e^{-\beta\varepsilon}}\right) = \text{diag}\left(\frac{1}{1 + e^{-1}}, \frac{e^{-1}}{1 + e^{-1}}\right),$$

and the benchmark values $I(\text{TFD}) = 2H(g)$ and $I(\text{deph}) = H(g)$ quoted later evaluate to 1.16441 and 0.582203 in nats.

3.6.1 LT geometry projection

Figure 1 shows a projection of the two-qubit LT body onto the three correlation coordinates

$$X_{xx} = \text{Tr}[C(\sigma_x \otimes \sigma_x)], \quad X_{yy} = \text{Tr}[C(\sigma_y \otimes \sigma_y)], \quad X_{zz} = \text{Tr}[C(\sigma_z \otimes \sigma_z)],$$

with $C = \rho - \Gamma$. Since $\text{Tr}_A C = \text{Tr}_B C = 0$, these axes isolate genuinely bipartite correlation content rather than local thermal bias. The origin is the product Gibbs state.

The figure should be read as a low-dimensional shadow of the 9-dimensional LT body, not as a literal cross-section. The compact central region corresponds to states well inside the positivity domain, while the outer shell is populated by boundary or near-boundary states. Analytic reference points for this body include the product Gibbs state at the origin, boundary TFD-type states, and the diagonal extremal states discussed earlier. The figure is therefore qualitative, but it still shows that LT is a substantial constrained body rather than a one-parameter family.

LT geometry: boundary + interior

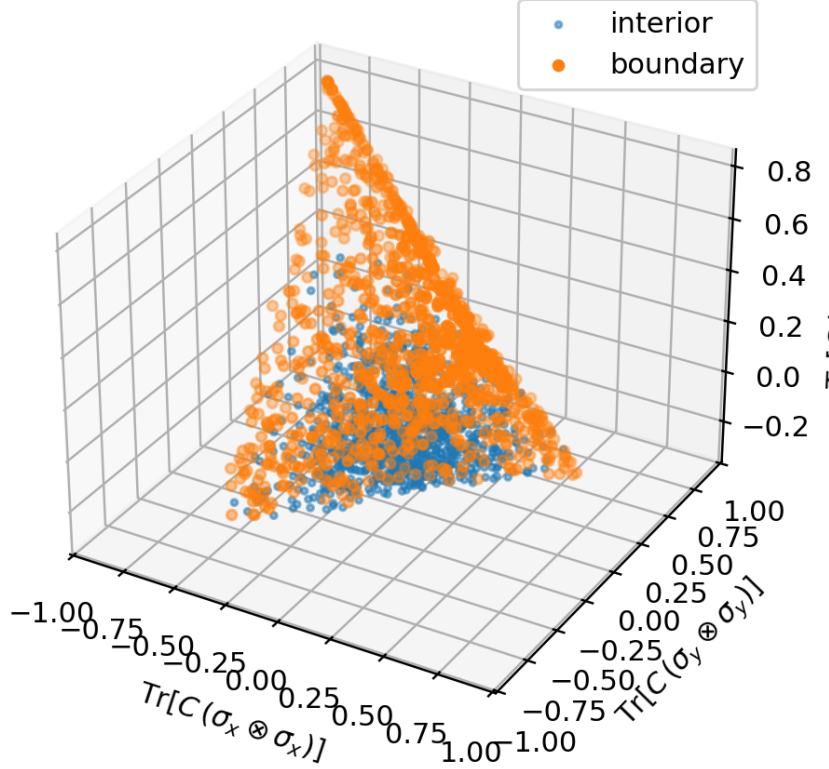


Figure 1: Projection of the two-qubit LT body onto the correlation coordinates X_{xx} , X_{yy} , and X_{zz} . The origin is the product Gibbs state $\Gamma = \gamma \otimes \gamma$.

The point cloud in Fig. 1 was generated by random sampling within the two-qubit LT parametrisation of Proposition 3.1. Concretely, candidate states were drawn by choosing the free LT parameters in the canonical form and retaining only those for which the resulting matrix is positive semidefinite. Each accepted state was then mapped to the three plotted coordinates

$$X_{xx} = \text{Tr}[C(\sigma_x \otimes \sigma_x)], \quad X_{yy} = \text{Tr}[C(\sigma_y \otimes \sigma_y)], \quad X_{zz} = \text{Tr}[C(\sigma_z \otimes \sigma_z)],$$

with $C = \rho - \Gamma$. The figure should therefore be read as an illustrative projection of feasible LT states, not as a uniform measure on the LT body or as an estimate of its volume. Its purpose is to visualise the shape of the allowed region and the location of representative boundary points such as Γ , TFD-type states, and diagonal extremal states.

3.6.2 Correlation equals athermality on LT

For the Pauli- XX family $\rho(p) = \Gamma + pC_{XX}$, Fig. 2 plots the relative entropy $D(\rho||\Gamma)$ and the mutual information $I(A:B)$. On LT one has the exact identity

$$D(\rho||\Gamma) = I(A:B),$$

so the two curves should coincide throughout the feasible interval. The figure is therefore mainly a consistency check and a visual reference for later convertibility plots rather than an independent

numerical discovery.

The common minimum at $p = 0$ identifies the product Gibbs state as the unique uncorrelated point on the ray, while the symmetry under $p \mapsto -p$ shows that opposite signs of the same XX correlation carry the same total correlation strength. Physically, once the local marginals are fixed, there is no independent local free-energy contribution left: the full thermodynamic resource relative to Γ is stored in correlations.

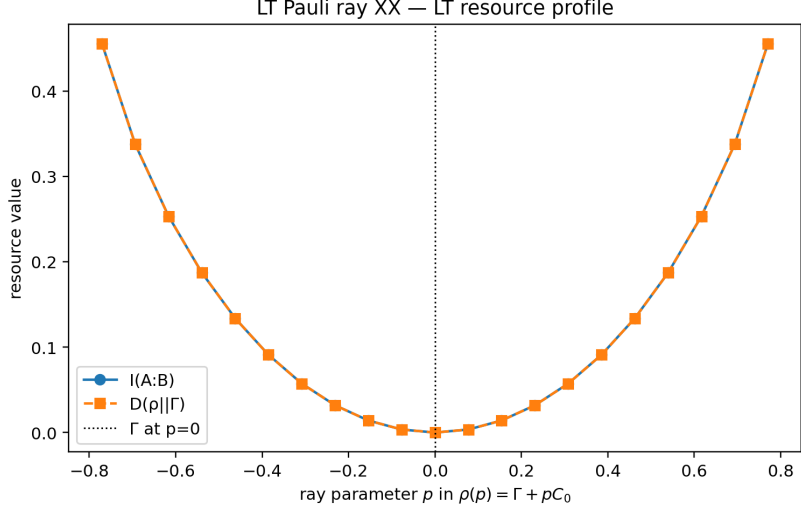


Figure 2: Resource profile along the Pauli- XX LT ray. The curves for $D(\rho||\Gamma)$ and $I(A : B)$ coincide throughout the feasible interval, as expected from the exact LT identity $D = I$.

3.6.3 Entanglement and PPT diagnostics on the LT ray

Along the diagonal- T LT ray, entanglement can be tracked using the PPT criterion and, when PPT fails, quantified by the negativity. In the two-qubit setting the Peres–Horodecki criterion is exact, so

$$\rho \text{ is separable} \iff \rho^{T_B} \succeq 0$$

[12, 13]. This makes PPT a sharp diagnostic for distinguishing the separable and entangled regions of the ray.

Numerically, the LT ray exhibits a central PPT sector around the product Gibbs state $\Gamma = \gamma \otimes \gamma$, with entangled behaviour appearing only after crossing the PPT boundary. The negativity is zero throughout the PPT region and becomes positive beyond it, so the ray gives a simple one-parameter illustration of how entanglement emerges inside the LT body while the local Gibbs marginals remain fixed. In this sense, the PPT boundary marks the transition from classically accessible correlation structure to a genuinely quantum LT sector.

3.6.4 Benchmark convertibility on a finite LT set

Before turning to the higher-resolution Pauli-XX ray, it is useful to look at a small benchmark family of eight representative LT states. This finite graph gives the cleanest coarse comparison between unrestricted global GP and verified local GP.

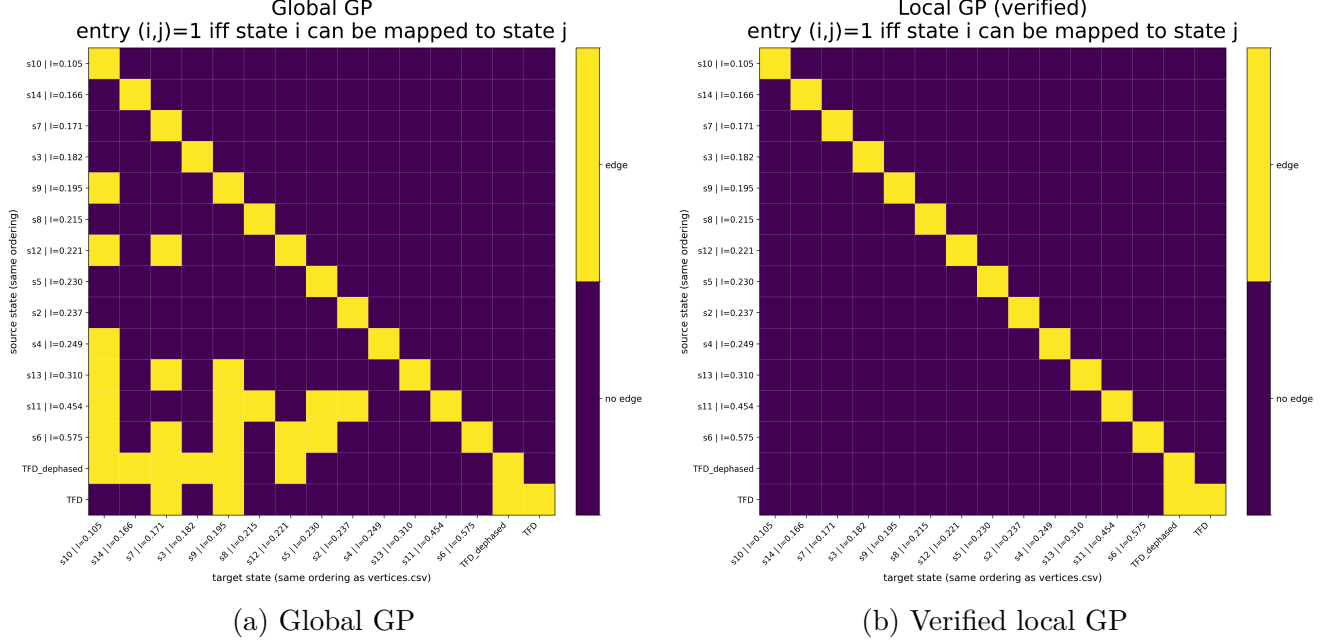


Figure 3: Convertibility adjacency matrices for a benchmark set of eight explicitly chosen LT states. An entry (i, j) is marked when the state in row i is found convertible to the state in column j under the class of maps shown in that panel. Left: global GP on the full bipartite system. Right: verified local GP, meaning product Gibbs-preserving channels.

The benchmark comparison already shows the main qualitative distinction between the two operational settings. Under global GP the graph contains several nontrivial edges, whereas under verified local GP almost all of these disappear. In the present benchmark data the adjacency counts are $N_{\text{global}} = 7$ and $N_{\text{local}} = 1$, so requiring the channel to factorise into local Gibbs-preserving maps makes convertibility much sparser on this finite LT set.

The cleanest positive benchmark edge is the transition from the thermofield-double state to its dephased LT counterpart. This conversion is feasible under both global and verified local GP, consistent with the analytic local-dephasing construction discussed earlier. Quantitatively, the resource decreases by exactly a factor of two:

$$I(\text{TFD}) = D(\text{TFD}||\Gamma) = 1.16441 \longrightarrow I(\text{deph}) = D(\text{deph}||\Gamma) = 0.582203.$$

This makes the TFD-to-dephased pair more than just one visible edge in the heatmap: it is the benchmark example showing that nontrivial correlation loss is still possible under local GP, but only in very special structured directions.

The two panels have a clear operational interpretation. Under global GP there are several ways to contract correlations while keeping the product Gibbs state fixed. Under verified local GP this freedom almost completely disappears once the map is required to factor as $\Phi_A \otimes \Phi_B$ with each

factor Gibbs-preserving.

This benchmark graph is deliberately small and is not intended as a complete description of LT convertibility. Its purpose is to show, on a controlled finite set, that local GP is markedly more restrictive than global GP. The Pauli XX heatmaps in the next subsection sharpen the same comparison on an ordered one-parameter LT family.

3.6.5 Convertibility heatmaps

Each convertibility heatmap is an adjacency matrix: an entry $(i, j) = 1$ means that the source state indexed by row i can be mapped to the target state indexed by column j by the class of maps shown in that panel. Here, global GP denotes unrestricted Gibbs-preserving maps on the full bipartite system, while PPT-outer denotes the PPT-based outer approximation introduced earlier.

For the sampled Pauli-XX LT family, the global and verified local Gibbs-preserving adjacency matrices were found to coincide on the sampled grid, so the local panel does not provide additional structure in this case. For that reason, the more informative comparison is between the global GP heatmap and the PPT-outer heatmap. States are ordered by increasing ray parameter p , so the matrices represent directed convertibility along a single physically transparent one-parameter LT family.

The global GP panel shows a broad connected convertibility region centred around the middle of the ray. This indicates that, within this family, many states can be mapped to other states with smaller or comparable correlation, producing the characteristic inward hourglass pattern. By contrast, the PPT-outer panel is much sparser and more fragmented. This shows that PPT positivity gives a much more restrictive criterion than actual Gibbs-preserving convertibility on this ray, and therefore serves only as a coarse outer approximation to the physically achievable transitions.

The coincidence of the local and global GP panels should be understood as a special feature of this highly symmetric one-parameter LT family, not as evidence that local and global Gibbs-preserving operations are equivalent in general.

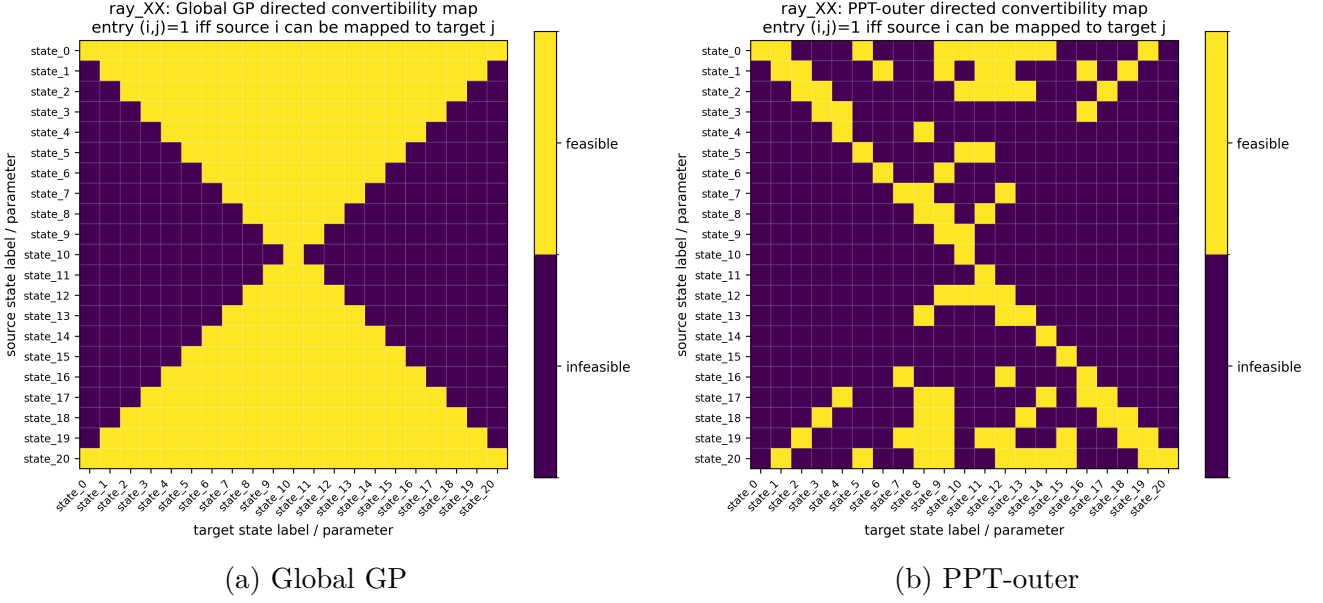


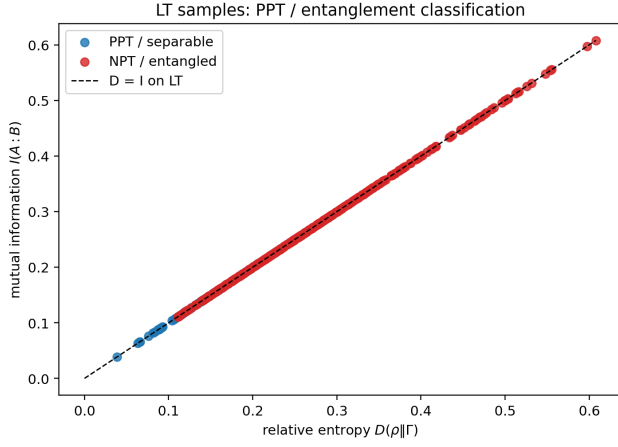
Figure 4: Convertibility heatmaps for the sampled Pauli-XX LT family, with states ordered by increasing ray parameter p . An entry (i, j) is marked when the state in row i is found convertible to the state in column j under the class of maps shown. On this family, the global Gibbs-preserving map yields a broad connected convertibility region, whereas the PPT-outer approximation is substantially sparser and more fragmented, showing that PPT provides only a loose outer bound on convertibility along this ray.

States are ordered by increasing correlation along the sampled family. For the Pauli-ray data this means increasing ray parameter p , which on LT also orders the common value $D(\rho||\Gamma) = I(A:B)$. For this, states are ordered by increasing mutual information $I(A:B)$.

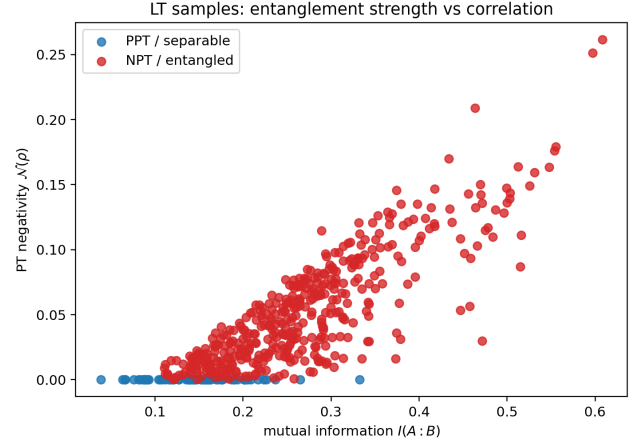
3.6.6 PPT and separability scatter results

The sampled LT ensemble provides a complementary statistical picture. In the $(D(\rho||\Gamma), I(A:B))$ plane, all points lie on the diagonal $D = I$, exactly as predicted on LT. For the particular numerical sampler used here, the PPT fraction is 0.113. This should be read only as a property of that sample, not as an intrinsic volume fraction of the LT set.

A second scatter plot, comparing mutual information with PT-negativity, sharpens the same conclusion. Weakly correlated states cluster near zero negativity, whereas larger correlation typically accompanies larger negativity. At the same time, states with $I(A:B) > 0$ but zero negativity remain visible, showing that LT contains a genuine separable sector with nontrivial correlation. In two qubits these PPT points are exactly the separable sector.



(a) $D(\rho||\Gamma)$ versus $I(A:B)$



(b) $I(A:B)$ versus PT-negativity

Figure 5: Scatter diagnostics for randomly sampled two-qubit LT states. Left: all points lie on the diagonal $D = I$. Right: stronger correlation generally accompanies larger PT-negativity, while PPT points remain confined to the low-correlation region.

Taken together, these plots show that LT states are correlated nonequilibrium states whose local thermodynamic content has already been fixed. Moving away from $\Gamma = \gamma \otimes \gamma$ therefore means building correlations rather than changing local marginals, exactly as captured by the LT identity $D(\rho||\Gamma) = I(A:B)$. The negativity scatter then refines this picture: separable LT states occupy a low-correlation region, while entangled LT states extend to larger correlation values. In this sense, entanglement provides a genuinely quantum advantage, enlarging the accessible correlation region beyond the classical/separable sector while keeping both subsystems locally thermal.

4 Discussion and conclusion

This project treats locally thermal states as a constrained state space whose geometry and operational behaviour can be studied together. Once both marginals are fixed to the same Gibbs state, all remaining freedom is global, so the problem becomes one of characterising the allowed correlation structure, understanding how the resulting LT families are organised, and determining how that hierarchy changes under different classes of Gibbs preserving dynamics.

For symmetric qubits, the report gives an explicit characterisation of the first fully nontrivial LT body. The canonical matrix form and Schur-complement positivity criterion describe the full geometry, while the diagonal T slice and LT rays provide tractable subfamilies that expose particular features of the wider set. For qutrits, by contrast, a full description is already much less tractable, so the commuting subclass becomes the natural analytic target. In that case the LT constraints reduce to a transportation polytope structure with an exact mutual information range. This naturally leads to a hierarchy picture. Starting from the product Gibbs state $\Gamma = \gamma \otimes \gamma$, one can move through commuting and separable LT families into coherent and entangled ones, always keeping the local thermal marginals fixed. The basic LT identity,

$$D(\rho||\Gamma) = I(A : B),$$

shows that global athermality above the product Gibbs baseline is exactly total correlation. Analytically this identity is immediate from the fixed marginals; numerically it serves mainly as a consistency check, while the PPT and scatter diagnostics show that entangled LT states reach a larger accessible correlation region than separable ones. In this sense, the quantum sector of the LT hierarchy supports stronger correlations than the classical or separable sector while preserving the same local Gibbs states.

Operationally, the hierarchy changes sharply depending on whether control is global or local. Global Gibbs preserving maps allow a substantial convertibility order toward less correlated states, whereas local Gibbs preserving maps are much more restrictive. The explicit TFD to dephased example shows that nontrivial local ordering does exist, but the numerical results indicate that verified local convertibility is far sparser than global GP. This distinction is one of the main conclusions of the report: LT is not only a geometric state space, but also an operational one whose ordering depends strongly on the allowed form of dynamics.

Several natural directions remain open. The first is higher dimensional characterisation, where the main challenge is to identify tractable LT subclasses and sharper boundary criteria beyond qubits. The second is to broaden the LT setting itself by allowing unequal local temperatures, asymmetric marginals, or different Hamiltonian structures. The third is operational: one would like stronger local convertibility tests, tighter SDP relaxations, and comparisons with more restrictive classes such as thermal operations or approximate transformations. Locally thermal state spaces connect questions from quantum thermodynamics, correlation geometry, and entanglement theory within a single constrained setting.

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Declaration concerning plagiarism

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Student number: 22336105

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Date: 22/03/2025

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A Appendix: proof sketches and implementation caveats

A.1 Proof sketches for the main structural results

Canonical two qubit LT form. Write a general Hermitian two qubit state in the product energy basis as

$$\rho = \begin{pmatrix} r_{00} & c_{01} & c_{02} & c_{03} \\ c_{01}^* & r_{01} & c_{12} & c_{13} \\ c_{02}^* & c_{12}^* & r_{10} & c_{23} \\ c_{03}^* & c_{13}^* & c_{23}^* & r_{11} \end{pmatrix},$$

with $r_{ij} \in \mathbb{R}$, $c_{ij} \in \mathbb{C}$, and $r_{00} + r_{01} + r_{10} + r_{11} = 1$. The marginal constraints

$$\mathrm{Tr}_{A'} \rho = \gamma, \quad \mathrm{Tr}_A \rho = \gamma$$

become

$$r_{00} + r_{01} = g, \quad r_{10} + r_{11} = 1 - g, \quad r_{00} + r_{10} = g, \quad r_{01} + r_{11} = 1 - g,$$

and

$$c_{02} + c_{13} = 0, \quad c_{01} + c_{23} = 0.$$

Among the four diagonal equations only three are independent because their sum is the trace constraint already imposed, so together with the two complex off-diagonal relations they give exactly six independent real constraints. Solving them gives

$$r_{01} = r_{10} =: a, \quad r_{00} = g - a, \quad r_{11} = 1 - g - a,$$

and, after the relabelling

$$c_{01} = x, \quad c_{02} = y, \quad c_{03} = z, \quad c_{12} = w,$$

one obtains

$$c_{13} = -y, \quad c_{23} = -x.$$

This yields precisely the canonical matrix form in Proposition 3.1. The free data are $a \in \mathbb{R}$ and $x, y, z, w \in \mathbb{C}$, hence $1 + 4 \times 2 = 9$ real parameters.

Schur-complement positivity theorem. For $\Delta > 0$, the block A in Theorem 3.2 is invertible. The standard block criterion gives $\rho \succeq 0$ if and only if $A \succeq 0$ and $D - B^\dagger A^{-1} B \succeq 0$. Substituting the explicit matrices yields the displayed formulas for N_{11} , N_{22} , and N_{12} ; in particular, the off-diagonal term uses w^* because it comes from $B^\dagger A^{-1} B$. Since the Schur complement is a Hermitian 2×2 matrix, its positivity is equivalent to nonnegative diagonal entries together with

nonnegative determinant.

Commuting qutrit mutual-information range. The lower bound follows from nonnegativity of relative entropy, $I(P) = D(P \| gg^T) \geq 0$, with equality if and only if $P = gg^T$. The upper bound follows from $I(X : Y) = H(X) - H(X|Y) \leq H(X) = H(g)$, with equality only when one variable is a deterministic function of the other. Equal marginals then force a permutation-type coupling.

TFD-to-dephased chain. Direct substitution gives $G_\mu(\rho_\lambda) = (1 - \lambda')\tilde{\gamma} + \lambda'\tilde{\gamma}_{\text{diag}}$ with $\lambda' = \lambda + \mu(1 - \lambda)$. Hence any pair $0 \leq \lambda \leq \lambda' \leq 1$ is connected by choosing $\mu = (\lambda' - \lambda)/(1 - \lambda)$.

A.2 Implementation

caveats

The experiment guide distinguishes three local-solver modes and states explicitly that `local_edge_mode=verified` is the mode to use for any figure or table intended for the report, whereas `single` is heuristic only. Accordingly, in the report the phrase *verified local GP* refers to edges returned by that verified mode; absence of a verified edge is not by itself a proof of impossibility. The main report keeps only validated numerical conclusions. Two outputs remain debugging targets rather than scientific claims. First, any reported failure of LT closure under genuinely local GP channels is inconsistent with the analytic closure theorem and therefore indicates an implementation problem, such as an error in the random channel generator, the Gibbs-preservation check, or the LT-membership tolerance logic. Second, if a PPT-on-joint-Choi test is being used as an outer relaxation of local GP, every genuinely verified local edge should also appear feasible in that relaxation up to common tolerances and consistent indexing. Failures of that inclusion should be treated as code-audit issues rather than physical effects.

A.3 Boundary

case $\Delta = 0$

In Theorem 3.2 the generic interior case was treated under the assumption that

$$\Delta := \det A = a\alpha - x^2 > 0,$$

so that A is invertible and the ordinary Schur complement applies. On the boundary $\Delta = 0$, the block A is positive semidefinite but singular, and one must use the singular Schur-complement criterion.

Write

$$\rho = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A = \begin{pmatrix} \alpha & x \\ x & a \end{pmatrix}, \quad B = \begin{pmatrix} y & z \\ w & -y \end{pmatrix}, \quad D = \begin{pmatrix} a & -x \\ -x & \delta \end{pmatrix}.$$

Then, for $A \succeq 0$ with $\Delta = 0$, one has

$$\rho \succeq 0 \iff A \succeq 0, \quad (I - AA^+)B = 0, \quad D - B^\dagger A^+ B \succeq 0,$$

where A^+ is the Moore–Penrose pseudoinverse.

In the gauge $x \geq 0$, the condition $\Delta = 0$ implies $x = \sqrt{a\alpha}$, so

$$A = uu^\dagger, \quad u = \begin{pmatrix} \sqrt{\alpha} \\ \sqrt{a} \end{pmatrix}, \quad A^+ = \frac{uu^\dagger}{(\alpha + a)^2}.$$

The range condition $(I - AA^+)B = 0$ means that each column of B must lie in the one-dimensional range of A . Equivalently,

$$\sqrt{a}y = \sqrt{\alpha}w, \quad \sqrt{a}z = -\sqrt{\alpha}y.$$

Once these compatibility conditions hold, positivity reduces to the 2×2 PSD condition

$$D - B^\dagger A^+ B \succeq 0.$$

Thus the boundary $\Delta = 0$ is a lower-rank compatibility locus: unlike the generic interior, the off-diagonal sector cannot vary freely but must align with the one-dimensional support of the singular block A .